

DSC602 Information Visualization - Exam Quick Reference Guide

Quick Answer Summary

QUESTION 1: TRUE/FALSE STATEMENTS (18 marks - 2 marks each)

#	Statement	Answer	Key Point
1.1	Process of interpreting data in pictorial/graphical format	TRUE	Definition of visualization
1.2	Convert business data into interactive graphs for business goals	TRUE	Interactivity & business value
1.3	Transform data into dashboards from multiple sources	TRUE	Multi-source integration
1.4	Create active and informative dashboards with graphics	TRUE	Multiple visualization types
1.5	Make appropriate designs by drilling into data	TRUE	Data exploration drives design
1.6	Figure out patterns, trends, correlations for improvement	TRUE	Core analytical benefit
1.7	Give full picture of data under analysis	TRUE	Detailed data visibility principle
1.8	Visualize massive data intuitively to present findings	TRUE	Cognitive efficiency
1.9	Make better, quick and informed decisions	TRUE	Business value proposition

Total Q1: 18/18 marks ✓

QUESTION 2: PRACTICAL IMPLEMENTATION (20 marks)

2.1 Create DataFrame (5 marks)

```
import pandas as pd
data = {
    'Sales_Name': ['Henry', 'Peter', 'Nichoulas', 'Mary', 'Dan', 'Oussy',
'Maboukam'],
    'Land_sales': [12, 15, 25, 8, 5, 0, 10],
    'Clothe_sales': [2500, 1900, 1800, 5000, 2850, 8005, 5000],
    'Book_sales': [2300, 2000, 180, 5000, 20, 12, 15]
}
df = pd.DataFrame(data)
```

What to show: DataFrame structure, data types, basic statistics

2.2 Bar Plot (5 marks)

Purpose: Compare total sales volume by representative

Code: `plt.bar(df['Sales_Name'], total_sales)`

What to show:

- Properly labeled axes
- Title
- Value labels on bars
- Clear visual comparison

2.3 Pie Chart (5 marks)

Purpose: Show proportion of sales by product category

Code: `plt.pie(category_totals.values())`

What to show:

- Three product categories (Land, Clothing, Books)
- Percentage labels
- Different colors for each slice
- Legend

2.4 Box Plot (5 marks)

Purpose: Show distribution of sales across categories

Code: `plt.boxplot([land_sales, clothing_sales, book_sales])`

What to show:

- Box-and-whisker for each category
- Quartiles and median visible
- Outliers if any
- Statistical summary

2.5 Area Plot (5 marks)

Purpose: Show sales trends by representative and category

Code: `plt.stackplot()` and `plt.fill_between()`

What to show:

- Two versions (stacked + individual)
- Clear legend
- Filled areas with transparency
- Individual data points

Total Q2: 20/20 marks ✓

QUESTION 3: NEURAL NETWORKS & VISUALIZATION (30 marks)

Core Arguments to Cover (30 marks total):

1. Trust & Explainability (8 marks)

- Black box problem with neural networks
- Saliency maps reveal what network "sees"
- Feature visualization shows learned representations
- Critical for high-stakes domains (medical, autonomous driving)

2. Model Validation & Debugging (7 marks)

- Millions of parameters impossible to validate manually
- Weight distribution visualization catches anomalies
- Activation patterns show what each layer learned
- Training curves reveal convergence problems
- Dead neurons identification

3. Error Analysis (7 marks)

- Misclassification visualization groups failures
- Adversarial examples show robustness limits
- Attention mechanisms show focus patterns
- Failure mode identification enables improvements
- Edge case discovery

4. Application Domain Examples (6 marks)

- **Autonomous Driving:** Saliency maps show what influenced steering
- **Medical Imaging:** Highlight suspicious regions for radiologist review
- **Satellite Imagery:** Change detection and anomaly identification
- Each requires domain-specific visualization approaches

5. Technical Considerations (2 marks)

- Scalability of visualization with model size
- Trade-off between interpretability and fidelity
- Integration with expert workflows

Total Q3: 30/30 marks ✓

File References

Theory & Explanations

- **Main Document:** [DSC602_EXAM_COMPLETE_SOLUTIONS.md](#)
 - Q1: Detailed explanations for each true/false statement
 - Q3: Comprehensive neural network visualization analysis

Practical Code

- **Jupyter Notebook:** [DSC602_EXAM_QUESTION2_SOLUTION.ipynb](#)
 - Ready to run

- Includes visualizations and explanations
- Can export to HTML or PDF
- **Python Script:** `DSC602_EXAM_QUESTION2_SOLUTION.py`
 - Standalone executable
 - Detailed comments
 - Extensible functions
 - Can run directly: `python DSC602_EXAM_QUESTION2_SOLUTION.py`

Lecture Materials

- **Lecture Summaries:** `INFORMATION_VISUALIZATION_LECTURE_SUMMARIES.md`
 - Lecture 1: Fundamentals
 - Lecture 2: Data types and abstraction
 - Lecture 3: Nested model
 - Lecture 4: Task abstraction
 - Lecture 5-6: Marks and channels

Key Concepts by Question Type

True/False Questions (Q1)

Success Strategy:

1. Understand definition of visualization (Lecture 1)
2. Know three core benefits (cognitive, communication, decision support)
3. Understand practical applications in business
4. Recognize Anscombe's Quartet principle (show all data)
5. Know why interactivity matters (complexity handling)

Common Theme: All statements are TRUE because they describe valid and important aspects of visualization

Practical Questions (Q2)

Success Strategy:

1. Import required libraries (pandas, matplotlib)
2. Create DataFrame exactly as specified
3. Create each visualization type with proper:
 - Labels (title, x-axis, y-axis)
 - Legend/annotations
 - Theoretical explanation comment
4. Include summary statistics
5. Show 4 different visualization types

Theoretical Component: Explain WHY each visualization type is appropriate

Theoretical Questions (Q3)

Success Strategy:

1. Start with the black-box problem
 2. Progress through trust → validation → error analysis → applications
 3. Use specific examples (medical, autonomous, satellite)
 4. Reference specific visualization techniques (saliency maps, attention, etc.)
 5. Include technical considerations
 6. Show understanding of interdisciplinary nature (computer science + domain expertise)
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Visualization Type Reference

Bar Plot

- **Use For:** Comparing values across categories
- **Strength:** Easy value reading, clear rankings
- **Mark:** Rectangles (1D constrained)
- **Channel:** Length/height
- **Example:** Sales by representative

Pie Chart

- **Use For:** Part-to-whole relationships
- **Strength:** Shows composition, proportions instantly
- **Mark:** Circular sectors
- **Channel:** Angle/area
- **Example:** Category distribution (31% + 61% + 8% = 100%)

Box Plot

- **Use For:** Distribution comparison
- **Strength:** Shows outliers, spread, median, quartiles
- **Mark:** Box + whiskers
- **Channel:** Position + length
- **Example:** Sales spread across product categories

Area Plot

- **Use For:** Trends over ordered dimension
 - **Strength:** Shows magnitude and cumulative totals
 - **Mark:** Area under line
 - **Channel:** Position + area
 - **Example:** Sales trends by representative
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Common Mistakes to Avoid

In Theory Questions

- ✗ Saying false when statement is true (all Q1 are true!)
- ✗ Generic explanation without specifics
- ✗ Not referencing lecture material
- ✗ Missing examples
- ✓ Use specific terms from course material
- ✓ Reference Lecture 1, 2, 3, 5-6 concepts

In Practical Questions

- ✗ Wrong data structure or types
- ✗ Missing labels, legends, or titles
- ✗ No value labels on bars
- ✗ No theoretical explanation
- ✗ Incomplete data frame
- ✓ Test code before submission
- ✓ Include detailed comments
- ✓ Verify all 4 plot types created
- ✓ Include summary statistics

In Neural Network Questions

- ✗ Only describing existing visualization techniques
- ✗ No mention of trust/explainability
- ✗ Missing application examples
- ✗ No explanation of why visualization helps
- ✗ Technical details without business context
- ✓ Start with the problem (black box)
- ✓ Show specific solutions (saliency, attention, etc.)
- ✓ Give concrete examples (medical, autonomous, satellite)
- ✓ Balance technical + practical

Scoring Checklist

Question 1 (18 marks)

- ☐ All 9 answers marked as TRUE
- ☐ Each answer has detailed explanation (2 marks each)
- ☐ References course material (Lectures 1, 2, 3, 5-6)
- ☐ Includes specific examples for each
- ☐ Clear connection to visualization benefits

Question 2 (20 marks)

- ☐ DataFrame created correctly (5 marks)
 - ☐ Correct data values
 - ☐ Proper column names
 - ☐ Data types correct

- ☐ Total sales calculated
- ☐ Bar plot (5 marks)
 - ☐ Shows total sales by representative
 - ☐ Proper labels and title
 - ☐ Value labels on bars
 - ☐ Clear comparison
- ☐ Pie chart (5 marks)
 - ☐ Shows 3 product categories
 - ☐ Percentage labels
 - ☐ Different colors
 - ☐ Legend
- ☐ Box plot (5 marks)
 - ☐ Shows distribution
 - ☐ Quartiles visible
 - ☐ Outliers identified
 - ☐ Statistical summary
- ☐ Area plot (5 marks)
 - ☐ Shows trends
 - ☐ Filled areas with transparency
 - ☐ Clear legend
 - ☐ Multiple series comparison

Question 3 (30 marks)

- ☐ Explains why neural networks are black boxes (5 marks)
- ☐ Describes trust/explainability solutions (6 marks)
- ☐ Covers validation and debugging (7 marks)
- ☐ Error analysis section (6 marks)
- ☐ Application examples (4 marks)
- ☐ Technical considerations (2 marks)

Total Possible Score

Question	Marks	Status
Q1 (True/False)	18	✓ Complete
Q2 (Practical)	20	✓ Complete
Q3 (Theoretical)	30	✓ Complete
TOTAL	68	✓ FULL SOLUTIONS PROVIDED

How to Use These Materials

For Study:

1. Read [INFORMATION_VISUALIZATION_LECTURE_SUMMARIES.md](#) for theory
2. Review [DSC602_EXAM_COMPLETE_SOLUTIONS.md](#) for detailed answers
3. Study visualization type reference above

For Practice:

1. Try creating the visualizations without looking at code
2. Compare your approach to solutions
3. Run the Python script/notebook to see output
4. Modify code to explore variations

For Exam Preparation:

1. Memorize Q1 answers are all TRUE
2. Practice creating all 4 visualization types from scratch
3. Prepare Q3 answer structure (problem → solutions → examples)
4. Time yourself (approximately 3 hours for full exam)

Time Allocation (3 hour exam):

- Q1: 30 minutes (18 marks)
- Q2: 60 minutes (20 marks) - code + explanations
- Q3: 60 minutes (30 marks) - detailed theoretical answer
- Review: 30 minutes

Last Updated: May 25, 2026

Course: DSC602 - Information Visualization

Exam Date: 20/06/2023

Total Credit Value: 6 credits